A Short Course in Analysis

Sample notes

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Chapter 1

Sequences

1.1 Convergence

Definition 1.1. *A sequence* (u_n) *of real numbers* converges *to* ℓ *if*

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N, \quad |u_n - \ell| < \varepsilon.$$

Example 1.2. *The sequence* $u_n = \dfrac{1}{n}$ *converges to* 0*: given* $\varepsilon > 0$*, take* $N > 1/\varepsilon$*.*

Theorem 1.3 (Monotone convergence)**.** *Every increasing sequence bounded above converges, to* $\sup_n u_n$*.*

Proof. Let $\ell = \sup_n u_n$. For $\varepsilon > 0$ there is N with $u_N > \ell - \varepsilon$, and then $\ell - \varepsilon < u_N \leq u_n \leq \ell$ for all $n \geq N$. □

1.2 Cauchy sequences

Definition 1.4. (u_n) *is a* Cauchy sequence *if* $\forall \varepsilon > 0,\ \exists N,\ \forall p, q \geq N,\ |u_p - u_q| < \varepsilon$*.*

Theorem 1.5. *In* $\mathbb{R}$*, a sequence converges if and only if it is a Cauchy sequence.*

Proof. A Cauchy sequence is bounded, so by Bolzano–Weierstrass it has a convergent subsequence $u_{\varphi(n)} \to \ell$; then

$$|u_n - \ell| \le |u_n - u_{\varphi(n)}| + |u_{\varphi(n)} - \ell| \longrightarrow 0. \qquad \square$$

Chapter 2

Series

2.1 Definitions

Definition 2.1. *The series $\sum u_n$ converges if its partial sums $S_N = \sum_{n=0}^{N} u_n$ converge.*

Example 2.2 (Geometric series)**.** *For $|q| < 1$,*

$$\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}.$$

Example 2.3 (Harmonic series)**.** $\displaystyle\sum_{n\geq 1} \frac{1}{n}$ *diverges, since $S_{2N} - S_N \geq \frac{1}{2}$.*

2.2 The Basel problem

Theorem 2.4 (Euler)**.**

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Idea. Compare the roots of $\dfrac{\sin x}{x} = \prod_{n\geq 1}\left(1 - \dfrac{x^2}{n^2\pi^2}\right)$ with its Taylor series $1 - \dfrac{x^2}{6} + \cdots$. □

Chapter 3

Continuity

3.1 Limits of functions

Definition 3.1. *f is continuous at a if* $\forall \varepsilon > 0,\ \exists \delta > 0,\ |x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$.

Theorem 3.2 (Intermediate value)**.** *If f is continuous on $[a, b]$ and $f(a) \leq y \leq f(b)$, there is $c \in [a, b]$ with $f(c) = y$.*

3.2 Uniform continuity

Theorem 3.3 (Heine)**.** *A continuous function on a closed bounded interval is uniformly continuous.*

Chapter 4

Integration

4.1 Riemann sums

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \frac{b-a}{n} \sum_{k=1}^{n} f\left(a + k\,\frac{b-a}{n}\right).$$

4.2 The Gaussian integral

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

Proof. Square it and switch to polar coordinates:

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy = \int_0^{2\pi}\int_0^{\infty} e^{-r^2}\,r\,dr\,d\theta = \pi. \qquad \square$$